

POLINOMIO DI TAYLOR

DATA UNA FUNZIONE f DERIVABILE n VOLTE IN x_0 , $\exists$ UNO ED UN SOLO POLINOMIO $T_n(x)$ DI GRADO $\leq n$ CON LA PROPRIETÀ CHE

$$T_n(x_0) = f(x_0), T_n'(x_0) = f'(x_0), \dots, T_n^{(n)}(x_0) = f^{(n)}(x_0)$$

TALE POLINOMIO SI CHIAMA **POLINOMIO DI TAYLOR** ED È

$$T_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

POLINOMIO DI CENTRO x_0 E GRADO n

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

SE $x_0 = 0$ $T_n(x)$ È DETTO POLINOMIO DI **MAC LAURIN** DI GRADO n

L'ERRORE CHE SI COMMETTE QUANDO SI APPROSSIMA $f(x)$ CON $T_n(x)$:

È DATO DA: $R_n(x) = f(x) - T_n(x)$
↓
RESTO

(A) FORMULA DI PEANO

↓

$$R_n(x) = o((x - x_0)^n) \quad \text{PER } x \rightarrow x_0 \quad \text{CIOÈ} \quad \lim_{x \rightarrow x_0} \frac{R_n(x)}{(x - x_0)^n} = 0$$

(B) FORMULA DI LAGRANGE

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SE f È DERIVABILE $n+1$ VOLTE IN $I(a, b)$ ESCLUSO AL PIÙ x_0

$\forall x \in I(a, b), \exists c$ COMPRESO TRA x E x_0 :

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

Sviluppi notevoli di McLaurin

$$(a) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^n}{(2n+1)!} x^{2n+1} + o(x^{2n+2})$$

$$(b) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n}{(2n)!} x^{2n} + o(x^{2n+1})$$

$$(c) e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + o(x^n)$$

$$(d) \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{(-1)^{n-1}}{n} x^n + o(x^n)$$

$$(e) \arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots + \frac{(-1)^n}{2n+1} x^{2n+1} + o(x^{2n+2})$$

$$(f) \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{1}{(2n+1)!} x^{2n+1} + o(x^{2n+2})$$

$$(g) \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{1}{(2n)!} x^{2n} + o(x^{2n+1})$$

$$(h) (1+x)^\alpha = 1 + \alpha x + \binom{\alpha}{2} x^2 + \binom{\alpha}{3} x^3 + \dots + \binom{\alpha}{n} x^n + o(x^n),$$

$$\text{dove } \binom{\alpha}{k} = \frac{\alpha \cdot (\alpha-1) \cdot \dots \cdot (\alpha-k+1)}{k!}$$

ESERCIZIO n°1

$$f(x) = \sin x \quad x_0 = 0 \quad f(0) = \sin(0) = 0$$

$$1. f'(x_0) = \cos x \Rightarrow f'(0) = \cos(0) = 1$$

$$2. f''(x_0) = -\sin x \Rightarrow f''(0) = -\sin(0) = 0$$

$$3. f'''(x_0) = -\cos x \Rightarrow f'''(0) = -\cos(0) = -1$$

$$4. f^{(4)}(x_0) = \sin x \Rightarrow f^{(4)}(0) = \sin(0) = 0$$

$$5. f^{(5)}(x_0) = \cos x \Rightarrow f^{(5)}(0) = \cos(0) = 1$$

$$T_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2} (x-x_0)^2$$

$$\left(\right. \\ T_n(x) = 0 + \underset{\substack{\uparrow \\ f'(x_0)}}{1} \frac{(x-0)^1}{1!} + \underset{\substack{\uparrow \\ f''(x_0)}}{0} \frac{(x-0)^2}{2!} + \underset{\substack{\uparrow \\ f'''(x_0)}}{-1} \frac{(x-0)^3}{3!} + \underset{\substack{\uparrow \\ f^{(4)}(x_0)}}{0} \frac{(x-0)^4}{4!} + \underset{\substack{\uparrow \\ f^{(5)}(x_0)}}{1} \frac{(x-0)^5}{5!} \Rightarrow x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

ESERCIZIO n°2

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \frac{\sin(0) - 0}{0^3} = \frac{0}{0} = 0 \quad \text{FORMA INDETERMINATA}$$

$$\lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{3!}\right) - x}{x^3} = \frac{-\frac{x^3}{3 \cdot 2}}{x^3} = \frac{-\frac{1}{6}x^3}{x^3} = -\frac{1}{6}$$

ESERCIZIO n°3

$$\ln(1+x) \quad x_0 = 0 \quad f(0) = \ln(1+0) = 0$$

$$f'(x_0) = \frac{1}{1+x} \Rightarrow f'(0) = \frac{1}{1+0} = 1$$

$$f''(x_0) = \frac{-1}{(1+x)^2} \Rightarrow f''(0) = \frac{-1}{(1+0)^2} = \frac{-1}{1+0^2} = -1$$

$$f'''(x_0) = \frac{2}{(1+x)^3} \Rightarrow f'''(0) = \frac{2}{(1+0)^3} = \frac{2}{1} = 2$$

$$T_n(x) = 0 + 1(x-0) + \frac{-1(x-0)^2}{2!} + \frac{2(x-0)^3}{3!}$$

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$$\ln(1+x) \approx x - \frac{1}{2!}x^2 + \frac{1}{3!}x^3$$

ESERCIZIO n°4

Sviluppo di TAYLOR

$$\lim_{x \rightarrow 0} \frac{3x - \sin 3x}{x(\ln(1+x) - x)} \quad x_0 =$$

$$n) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

$$n) \sin 3x = 3x - \frac{(3x)^3}{3!} + \frac{(5x)^5}{5!} + o(x)^3$$

$$d) \ln(1+x) = x - \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + o(x)^3$$

$$\lim_{x \rightarrow 0} \frac{3x - \sin 3x}{x(\ln(1+x) - x)} = \frac{\cancel{3x} - \left(\cancel{3x} - \frac{(3x)^3}{3!} + \frac{(5x)^5}{5!}\right)}{x\left(\cancel{x} - \frac{1}{2!}x^2 + \frac{1}{3!}x^3 - \cancel{x}\right)} = \frac{\frac{(3x)^3}{3!}}{x\left(\frac{1}{2!}x^2 + \frac{x^3}{3!}\right)} \Rightarrow \frac{\frac{27}{3 \cdot 2}x^3}{\frac{1}{2}x^3 + \frac{x^4}{3}} = \frac{\frac{9}{2}x^3}{\frac{1}{2}x^3 + \frac{x^4}{3}} = \frac{\cancel{\frac{9}{2}}x^3}{\cancel{\frac{1}{2}}x^3 + \frac{\cancel{2}}{3}x^4}$$

ESERCIZIO n°5

Sviluppo di TAYLOR

$$\lim_{x \rightarrow 0} \frac{\sin(e^x - 1) - x + \frac{x^2}{2}}{x^4}$$

- $e^x - 1 = x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4$
- $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!}$
- $\sin(e^x - 1) = x + \frac{1}{2}x^2 - \frac{5}{24}x^4$

$$f(0) = \sin(e^0 - 1) = 0$$

$$f'(x_0) = \cos(e^x - 1)e^x \Rightarrow f'(0) = \cos(e^0 - 1)e^0 = 1$$

$$f''(x_0) = -\sin(e^x - 1)e^{2x} + \cos(e^x - 1)e^x =$$

$$f''(0) = -\sin(e^0 - 1)e^{2 \cdot 0} + \cos(e^0 - 1)e^0 = 1$$

$$f'''(x_0) = -\cos(e^x - 1)e^{3x} - 3e^{2x} \cdot \sin(e^x - 1) + \cos(e^x - 1) \cdot e^x$$

$$f'''(0) = -\cos(e^0 - 1)e^{3 \cdot 0} - 3e^{2 \cdot 0} \cdot \sin(e^0 - 1) + \cos(e^0 - 1) \cdot e^0 = 0$$

$$f^{(4)}(x_0) = \sin(e^x - 1)e^{4x} - 6e^{3x} \cdot \cos(e^x - 1) - 7e^{2x} \cdot \sin(e^x - 1) + \cos(e^x - 1)e^x$$

$$f^{(4)}(0) = \sin(e^0 - 1)e^{4 \cdot 0} - 6e^{3 \cdot 0} \cdot \cos(e^0 - 1) - 7e^{2 \cdot 0} \cdot \sin(e^0 - 1) + \cos(e^0 - 1)e^0 = -6 + 1 = -5$$

$$T_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

$$T_4(x) = 0 + 1(x - 0) + \frac{1}{2!}(x - 0)^2 + \frac{0}{3!}(x - 0)^3 + \frac{-5}{4!}(x - 0)^4$$

4 FATORIAL 4!

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

$$T_4(x) = x + \frac{1}{2}x^2 - \frac{5}{24}x^4$$

$$\sin(e^x - 1) = x + \frac{1}{2}x^2 - \frac{5}{24}x^4$$

$$\lim_{x \rightarrow 0} \frac{\sin(e^x - 1) - x + \frac{x^2}{2}}{x^4} = \frac{(x + \frac{1}{2}x^2 - \frac{5}{24}x^4) - x + \frac{1}{2}x^2}{x^4} = \frac{-\frac{5}{24}x^4}{x^4} = -\frac{5}{24}$$